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CMSE 801 - Day 11

Testing, modifying, and
visualizing compartmental
models

Feb 17, 2026



General Course Announcements

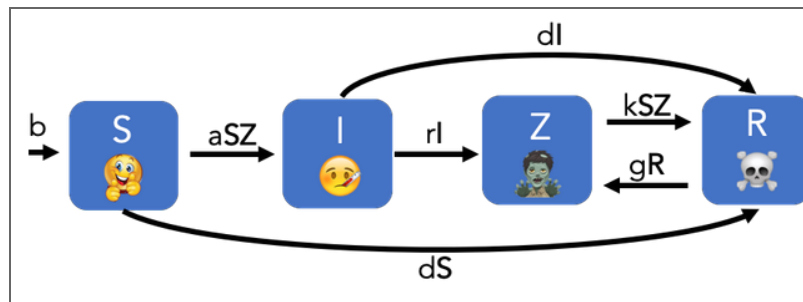
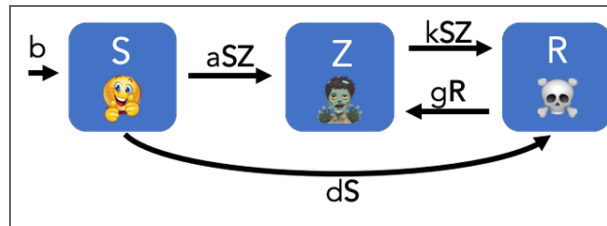
- **Reminder:** Homework 2 is on the course website and is due by **11:59 on Saturday, Feb 28.**
- I'll be working on grading your quizzes this week.

Challenges from Day 11 PCA

- Figuring out how to modify the models
- plotting
- SIZR/SIZRC

SZR → SIZR Model

I = infected



In []:

```
def SIZR(t, y, b, d, a, g, k, r):
    Si = y[0]
    Ii = y[1]
    Zi = y[2]
    Ri = y[3]
    # the model equations (see Munz et al. 2009)
    f0 = b - a*Si*Zi - d*Si
```

```
f1 = a*Si*Zi - r*Ii - d*Ii  
f2 = r*Ii + g*Ri - k*Si*Zi  
f3 = d*Si + d*Ii + k*Si*Zi - g*Ri  
return [f0, f1, f2, f3]
```

Goals for today

- For the model your group is assigned, you'll run the basic compartmental model and then explore the properties of the model and how changes to the model influence the results.
- As a group, you'll want to think about how you can modify or enhance the model, try to implement those ideas, and study the impact of those changes.
- Collectively, you'll create a slide or two that showcases your model and presents your findings
 - We'll try to go around the room at the end of class and each group will present (as time permits)

Looking forward - Day 12 PCA: A reminder/review of computational models and exploring data with the Pandas data analysis library

- Pandas is the most commonly used data analysis package/library in Python and will be the primary tool we use for doing data analysis in the course
- You'll explore some of the basics of pandas
- You'll spend a bit of time reflecting on what we mean when we talk about scientific models and computational models -- we've been working with models already in class, but it's good to pause and make sure you know what we mean when we talk about model!

In []:

